

3.1 Technical principles

3.1.1 Materials and their applications

Materials and applications

Content	Potential links to maths and science
Students are expected to be able to name specific materials for a wide range of applications. They must also be able to provide detailed and justified explanations of why specific materials and combinations of materials are suitable for given applications with reference to: • physical and mechanical properties and working characteristics • product function • aesthetics • cost • manufacture and disposal.	Understand the appropriate use of textile materials based on their physical and working characteristics such as: • thermal insulation • ability to reflect light • thermoplasticity • flammability • ability to absorb moisture • development of static electricity. Calculation of quantities of materials sizes and costs.

Classification of materials

Content	Potential links to maths and science
Students must know the classifications of the following fibres, and be able to name examples that belong to each category: • natural fibres: • natural plant/cellulosic fibres – cotton, linen, ramie • natural animal/protein fibres – wool, silk • hair fibres – cashmere, mohair, angora • manufactured fibres: • regenerated fibres – viscose, acetate • new generation lyocells – Modal®, Tencel®, Lyocell® • synthetic fibres: • polyamide (nylon), polyester, acrylic, elastomeric • chlorofibres (polyvinyl), fluorofibres (PTFE) • aramid fibres (Kevlar®, Nomex®) • inorganic fibres including glass, carbon, metallic, ceramic • smart materials: • reactive materials • photochromic dyes • phase changing materials • modern materials: • microfibres • nano-fibres • microencapsulated fibres and fabric.	

Methods for investigating and testing materials

Content	Potential links to maths and science
Students must understand how workshop and industrial tests are set up and what will be tested, measured and compared, including: • flammability • crease resistance • shrink resistance • colour fastness • strength • pilling.	Analysis of data obtained from testing.